

Ayu Dewi - Epidemiology and Data Science department

Open Science & Reproducibility

R-USER MEETING - AMSTERDAM UMC

28 May 2026

Rabies in Bali



Designing Programs for Eliminating Canine Rabies from Islands: Bali, Indonesia as a Case Study

Sunny E Townsend*, I Putu Sumantra, Pudjiatmoko, Gusti Ngurah Bagus, Eric Brum, Sarah Cleaveland, Sally Crafter, [A.P.M. Dewi](#), Dewa Made Ngurah Dharma, Jonathan Dushoff, Janice Girardi, I Ketut Gunata, Elly F Hiby, Corlevin Kalalo, Darryn L Knobel, I Wayan Mardiana, AA Gde Putra, Luuk Schoonman, Helen Scott-Orr, Mike Shand [Show 4 more](#)

*Corresponding author for this work

Boyd Orr Centre for Population and Ecosystem Health, Institute for Biodiversity, Animal Health and Comparative Medicine, College of Medical, Veterinary and Life Sciences, University of Glasgow, Glasgow, Scotland, United Kingdom, Bali Province Livestock Services, Jalan Asoka no. 14, Kreneng, Denpasar, Bali, Indonesia, Directorate General of Livestock and Animal Health Services, Ministry of Agriculture of the Republic of Indonesia, Jalan Harsono, Jakarta, Indonesia, Bali Animal Welfare Association, Ubud, Bali, Indonesia, Food and Agriculture Organization (FAO), Emergency Centre for Transboundary Animal Diseases (ECTAD), Jakarta, Indonesia, Disease Investigation Center Denpasar, Denpasar, Bali, Indonesia, Department of Biology, McMaster University, Hamilton, Ontario, Canada, Badung District Livestock Services, Jalan Raya Sempidi, Mangupura, Badung, Bali, Indonesia, World Society for the Protection of Animals, London, United Kingdom, Department of Veterinary Tropical Diseases, Faculty of Veterinary Science, University of Pretoria, Onderstepoort, South Africa, Faculty of Veterinary Science, University of Sydney, Camden, New South Wales, Australia, School of Geographical and Earth Sciences, University of Glasgow, Glasgow, Scotland, United Kingdom

Research output: Contribution to journal > Article > Academic > peer-review

The outbreak began in **2008** with highest mortality cases in **2011**.

Rabies elimination program was conducted.

By **2013....** mortality rates reduced:



98 % - humans

70% - dogs



After a major success, the rabies elimination programmes eventually came to an end.

Researchers left the programme.

However...

The data and analysis code were never shared, because they were considered confidential (due to data ownership issue).



The screenshot shows the Amsterdam UMC website. At the top is the logo and name 'Amsterdam UMC'. Below it is a search bar. A navigation menu includes 'Home', 'Researchers', 'Departments & Research Institutes', 'Projects & Funding', 'Publications' (which is underlined), and 'Datasets'. Below the navigation menu are links for 'Press/Media' and 'Impact'. The main content area features the title 'Effects of Rabies Elimination Program on Rabies Cases in Bali, 2008–2015' in a large, bold, dark blue font. Below the title, the authors 'A.P.M. Dewi*', 'Pandu Riono', and 'Muhammad Noor Farid' are listed. A note indicates that A.P.M. Dewi is the corresponding author. The affiliation 'Faculty of Public Health, Universitas Indonesia, Depok, Indonesia' is provided. At the bottom, a breadcrumb trail reads 'Research output: Contribution to journal > Article > Academic > peer-review'.

Amsterdam UMC

Search...

Home Researchers Departments & Research Institutes Projects & Funding Publications Datasets

Press/Media Impact

Effects of Rabies Elimination Program on Rabies Cases in Bali, 2008–2015

A.P.M. Dewi*, Pandu Riono, Muhammad Noor Farid

*Corresponding author for this work

Faculty of Public Health, Universitas Indonesia, Depok, Indonesia

Research output: Contribution to journal > Article > Academic > peer-review

The programmes were continued by a new group of researchers, but in 2014, news reports described chaos and human deaths caused by rabid dogs.

One of the former researchers (aka me) investigated this...

Findings:

The new researchers **did not use exactly the same “recipes”** as the previous researchers had used.

There was a **400% increase in mortality** cases in humans and dogs compared to the reports from 2011.



This is just a few of examples where non-reproducible research can lead to **ineffective health outcomes**.

The story continues...

Funders study reproducibility (28 funders)



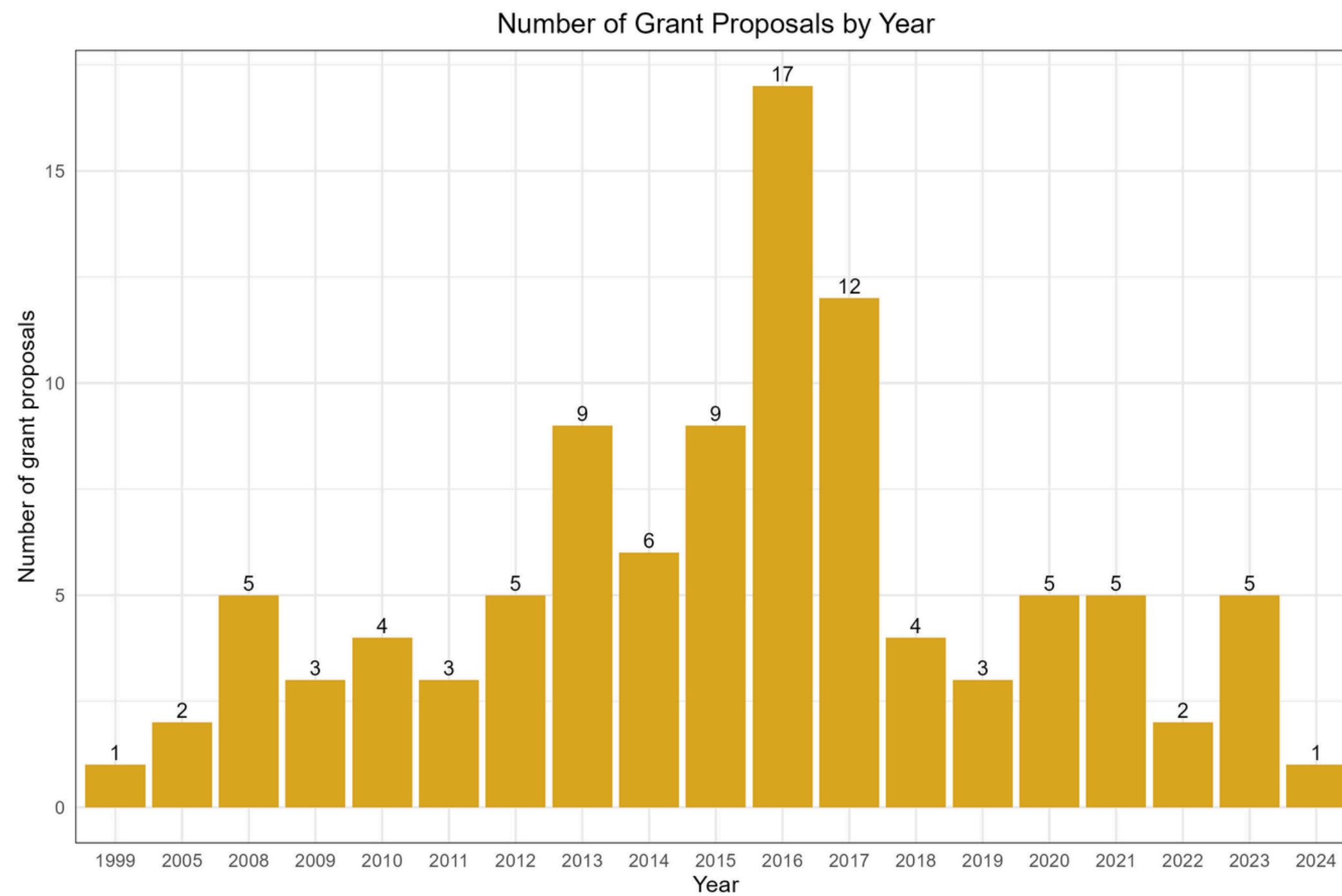
Marie Skłodowska-Curie



novo nordisk
foundation



Link to preprint: https://osf.io/preprints/metaarxiv/dvhux_v1text



Link to preprint: https://osf.io/preprints/metaarxiv/dvhux_v1text

Only 7% publication funded by these major grants are reproducible.





WHY ?

Let's dive in how reproducibility checks we had.....

Reproducible = original finding matched with the reproduction, using same data and same analytical methods

Original

Table 3. Number of ecosystem services' evaluations by continent and percentage total.

Continent	No of evaluations	% of total evaluations
Europe	259	50
Asia	160	31
North America	33	6
Africa	31	6
South America	26	5
Oceania	7	1

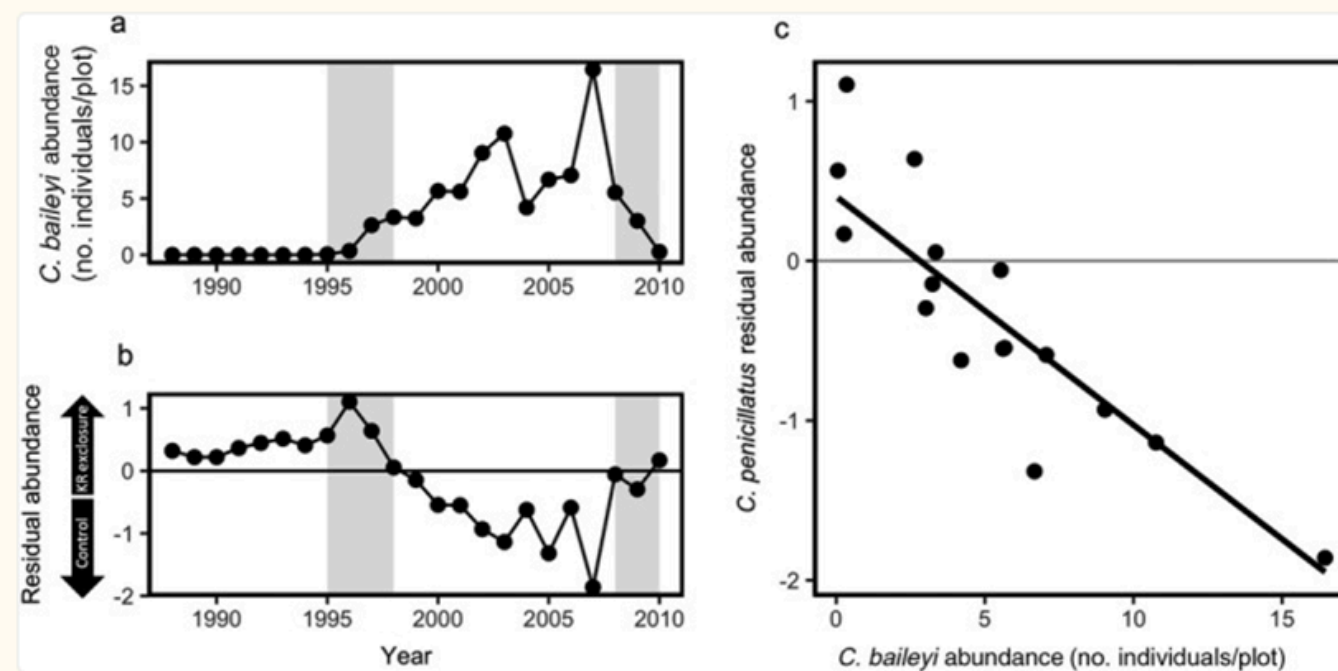
Manuscript: <https://doi.org/10.1016/j.ecolind.2024.112310>

Reproduction

Continent	Total
Africa	31
Asia	161
Europe	213
Global (All)	2
North America	33
Oceania	7
South America	26
World	1
	1
Grand Total	475

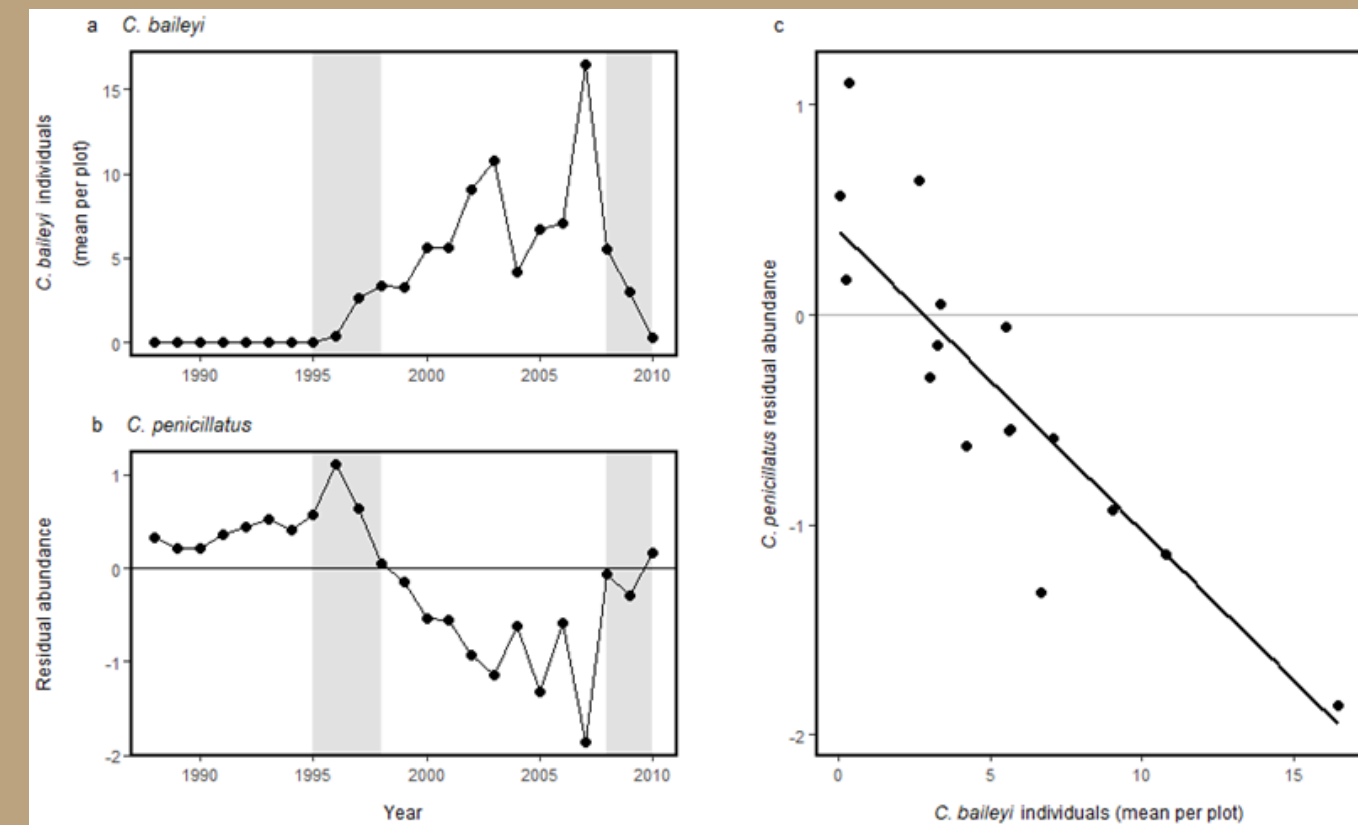
Original

Figure 1.



<https://pmc.ncbi.nlm.nih.gov/articles/PMC6899716/>

Reproduction

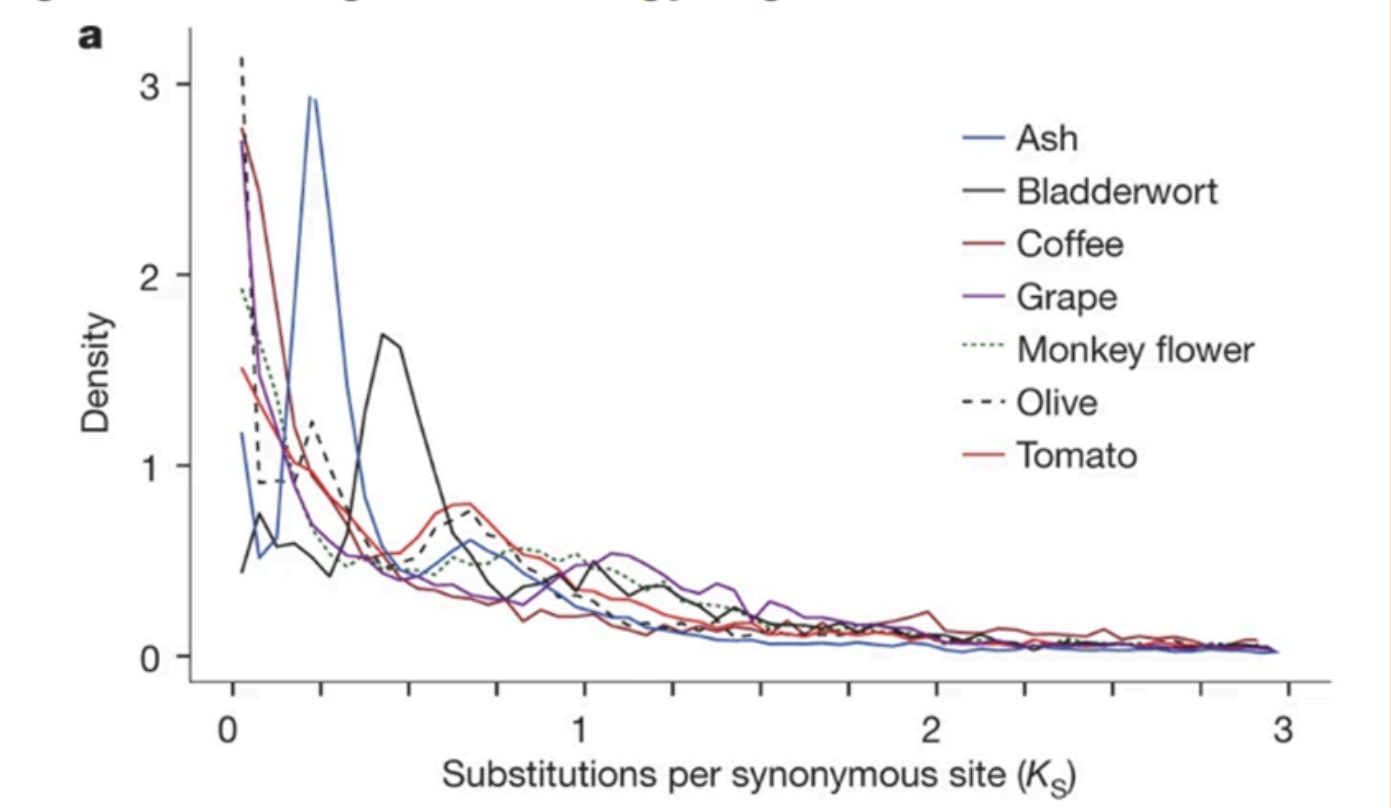


Data availability

The reference tree is growing at Earth Trust with accession number 2451S. Trimmed DNA and RNA reads and the final assembly for the 2451S genome sequence, as well as RNA reads for parent tree and raw reads and consensus read mappings of the European diversity panel trees, have been deposited in European Nucleotide Archive under project accession code PRJEB4958 (<http://www.ebi.ac.uk/ena/data/view/PRJEB4958>). Metabolomic data that support the findings of this study have been deposited in MetaboLights under accession code MTBLS372 (<http://www.ebi.ac.uk/metabolights/MTBLS372>). All other data are available from the corresponding author upon reasonable request.

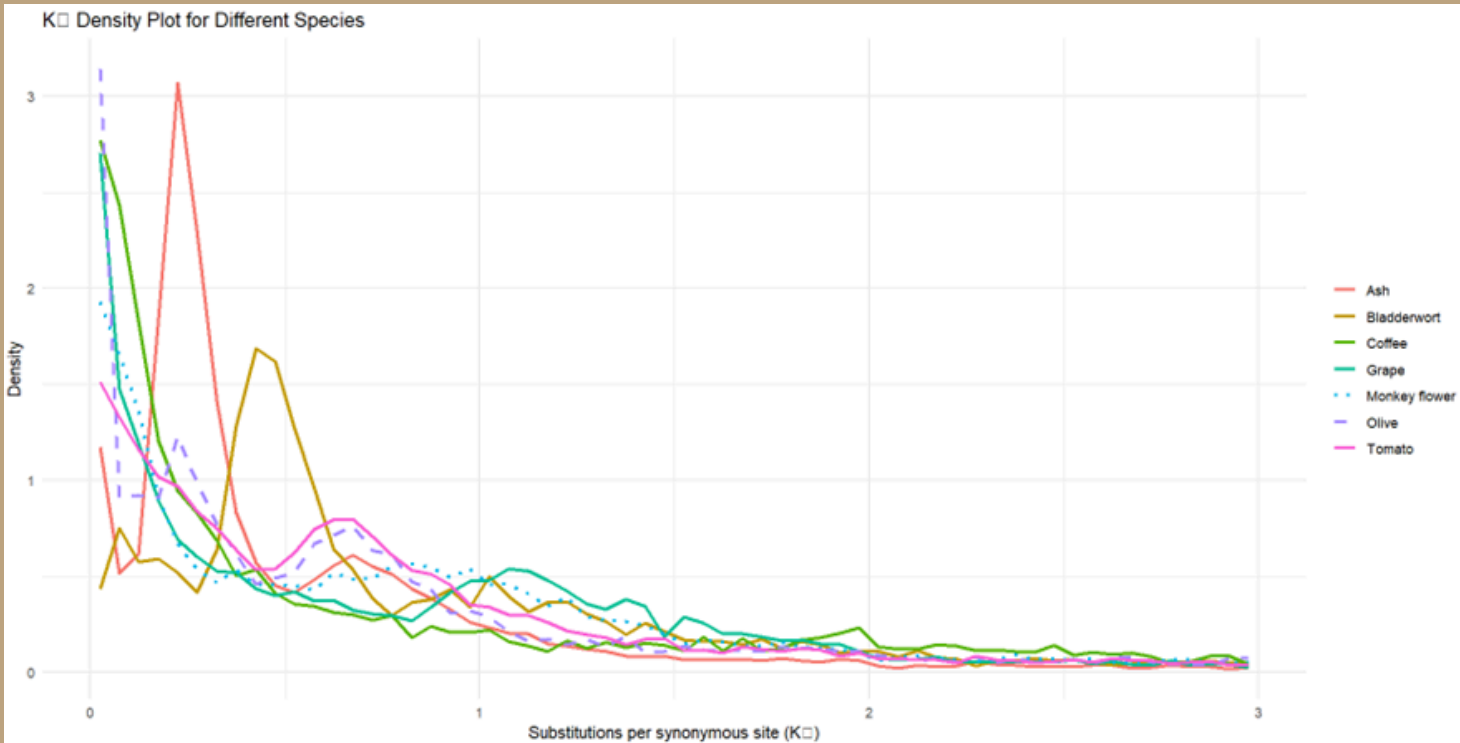
Original

Figure 1: Gene sharing within and among plant genomes.



<https://www.nature.com/articles/nature20786>

Reproduction



Original

Table 3 Summary of the mixed effects models to estimate log ρ , and the log ratios of competitive Effects and Responses. “Source” are random effects associated with individual studies. “Var” is variance.

	log ρ	Effects	Responses
Intercept	−1.57	−1.40	−1.31
[95% C.I.]	[−2.24; −0.86]*	[−2.00; −0.75]*	[−1.79; −0.79]*
Num. obs.	136	384	527
Num. groups:	21	27	33
Source			
Var: Source	1.37	1.07	1.05
Var: Residual	4.06	6.10	5.47
Num. groups:	—	86	200
Species:Source			
Var: Species:Source	—	1.24	0.44

*0 outside the confidence interval.

<https://doi.org/10.1111/ele.12793>

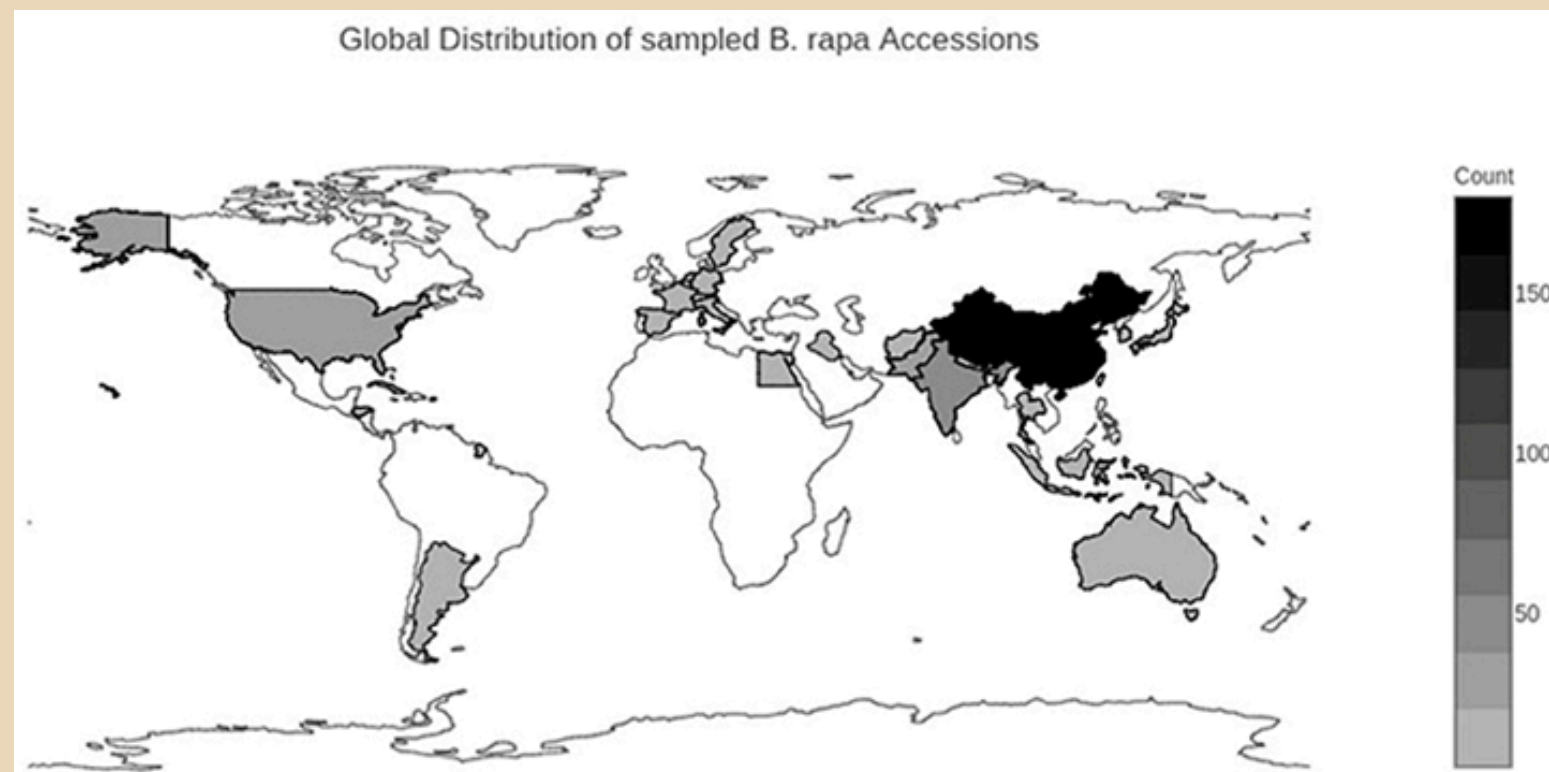
Reproduction

Summary of the mixed effect models

	Rho	Effects	Responses
(Intercept)	−1.57 [−2.24; −0.86]	−1.40 [−2.00; −0.75]	−1.31 [−1.79; −0.79]
AIC	602.40	1858.23	2463.88
BIC	611.14	1874.03	2480.95
Log Likelihood	−298.20	−925.12	−1227.94
Num. obs.	136	384	527
Num. groups: Source	21	27	33
Var: Source (Intercept)	1.37	1.07	1.05
Var: Residual	4.06	6.10	5.47
Num. groups: Intra.species:Source		86	
Var: Intra.species:Source (Intercept)		1.24	
Num. groups: Target.species:Source			200
Var: Target.species:Source (Intercept)			0.44

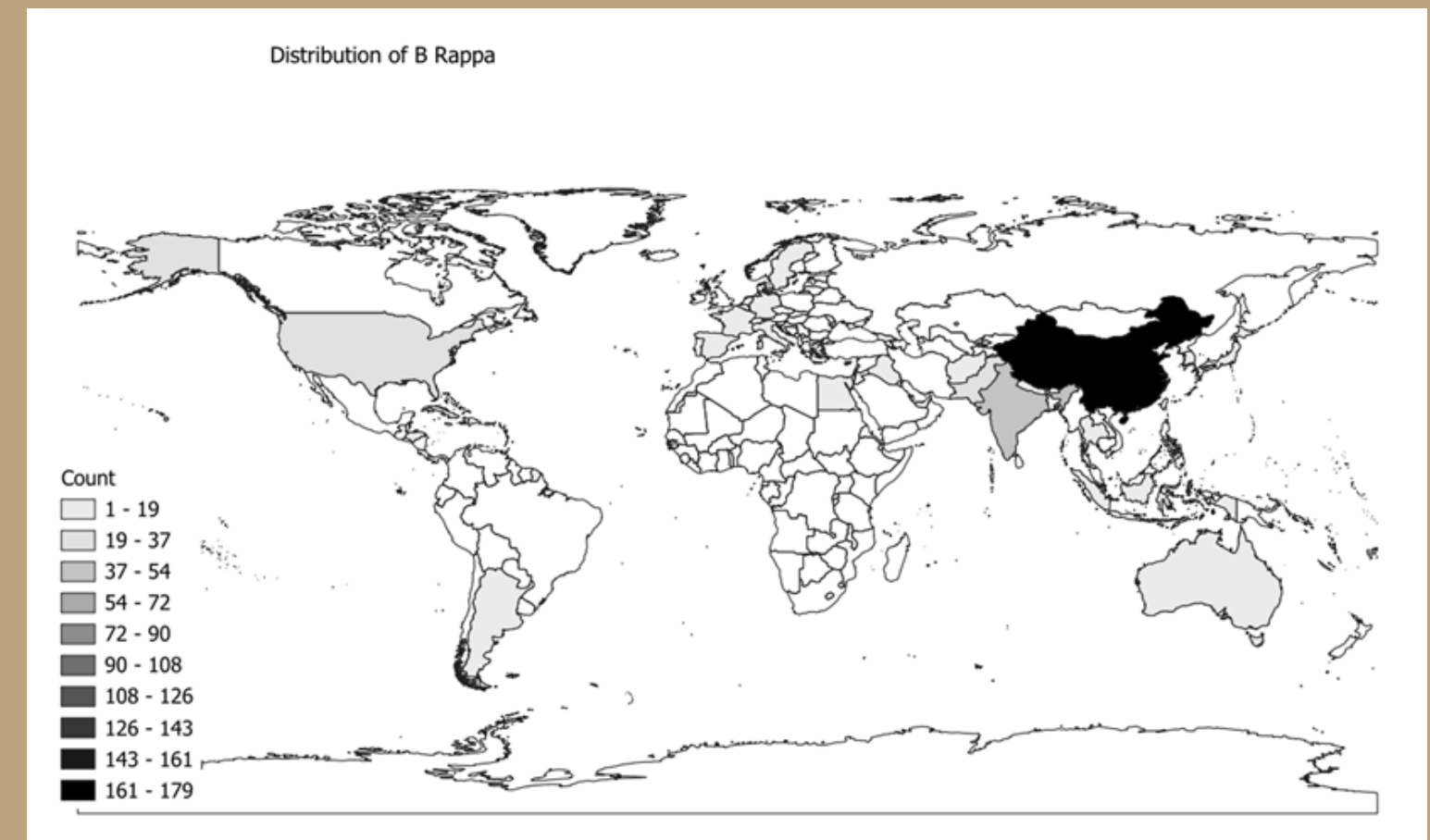
* Null hypothesis value outside the confidence interval.

Original

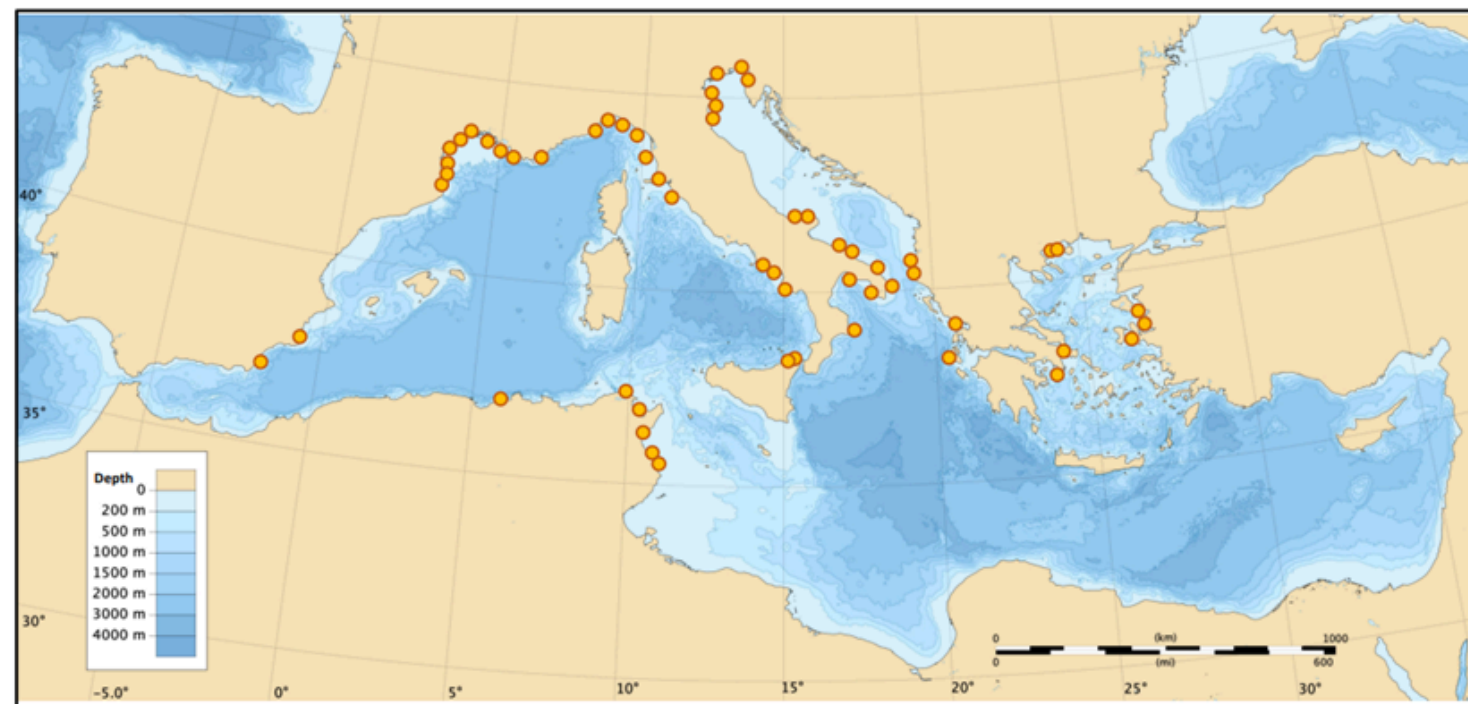


Manuscript: <https://doi.org/10.3389/fpls.2017.00321>

Reproduction



Original

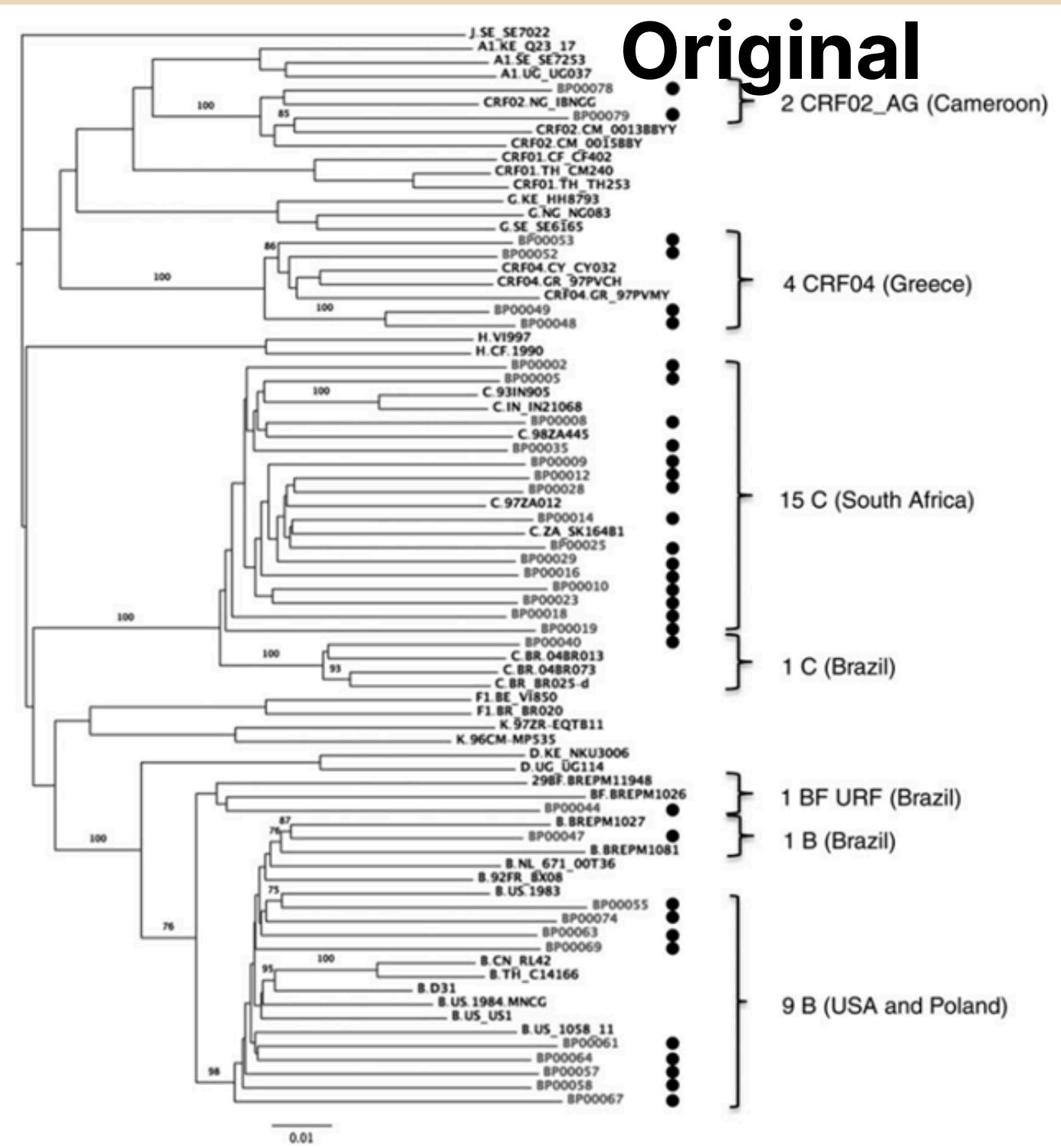


Map of sites with EQR for coastal soft bottoms quantified using M-AMBI.

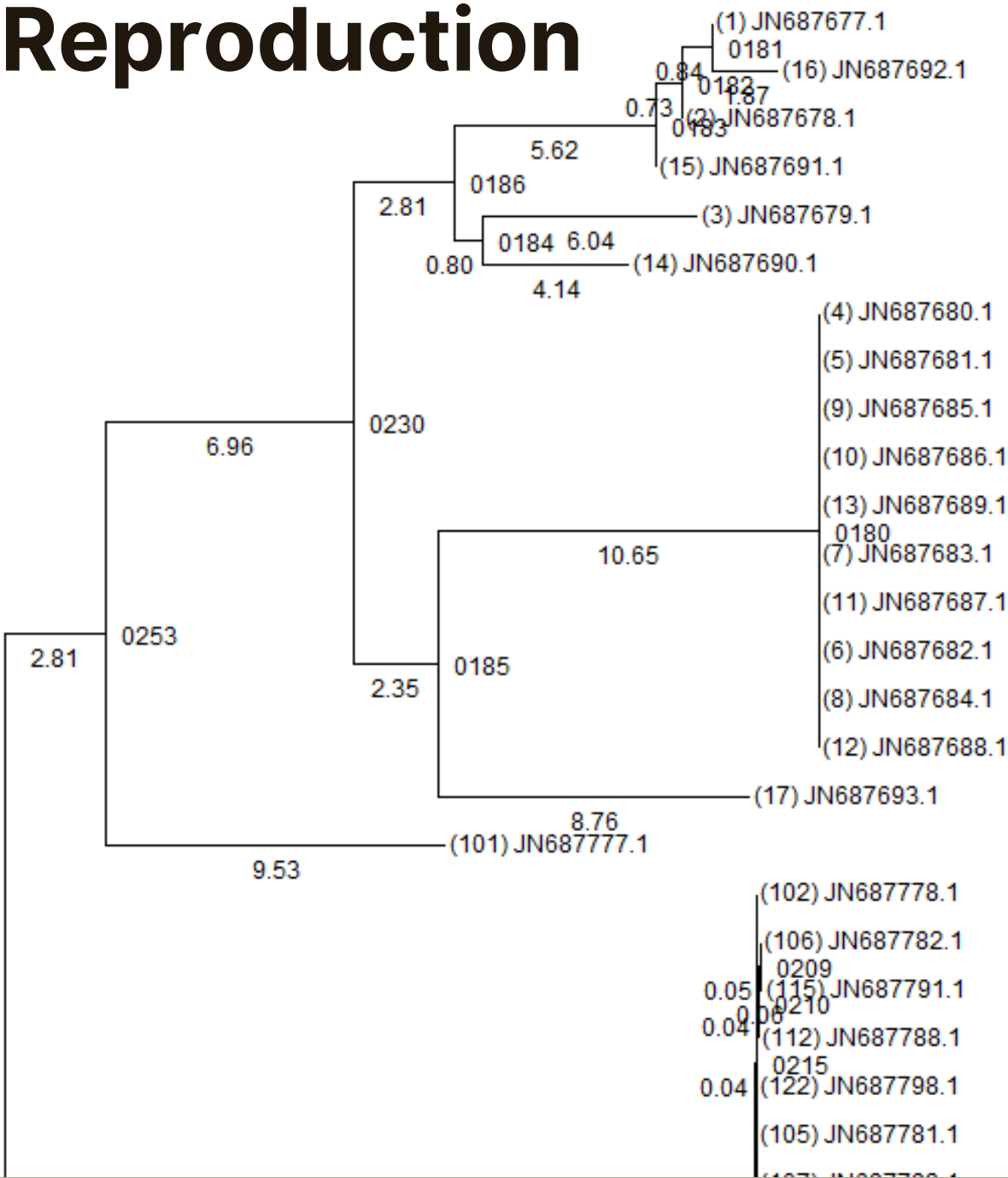
Manuscript: <https://doi.org/10.3389/fmars.2020.00475>

Reproduction





Reproduction



MA) and sequenced by an Applied BioSystems 3130/3130xl Genetic Analyzer. Sequences were assembled using Sequencher, version 4.7, and sequence alignments visualized using Highlighter (www.hiv.lanl.gov). Sequences were aligned using Clustal and phylogenetic analyses were performed using the PHYLIP package. A neighbor-joining tree of sequences of interest and reference strains was constructed. Diversity metrics were obtained using DIVEIN (<http://indra.mullins.microbiol.washington.edu/DIVEIN/>).³¹ Plasma- and culture-derived *env* sequences were analyzed using Poisson-Fitter (http://www.hiv.lanl.gov/content/sequence/POISSON_FITTER/poisson_fitter.html).³² Recombinants were analyzed using the jumping profile hidden Markov model (jpHMM) tool provided on www.hiv.lanl.gov.

CLUSTAL
Neighbor-joining tree

Sequence Data

Sequences from the isolates described in this study have been submitted to GenBank (www.ncbi.nlm.nih.gov/genbank/) under accession numbers JN687677 to JN687847.

NIH National Library of Medicine
National Center for Biotechnology Information

Nucleotide Nucleotide Search

Species Viruses (171) Customize ...

Molecule types genomic DNA/RNA (171) Customize ...

Source databases INSDC (GenBank) (171) Customize ...

Summary 200 per page Sort by Default order

Items: 171

1. [HIV-1 isolate BP00061_LH03 from USA gag protein \(gag\) gene, complete cds; vif protein \(vif\) and vpr protein \(vpr\) genes, complete cds; and pol protein \(pol\) gene, partial cds](#)

5,063 bp linear RNA

Accession: JN687677.1 GI: 363584084

[Protein](#) [PubMed](#) [Taxonomy](#)

[GenBank](#) [FASTA](#) [Graphics](#)

2. [HIV-1 isolate BP00063_LH01 from USA gag protein \(gag\) gene, complete cds; vif protein \(vif\) and vpr protein \(vpr\) genes, complete cds; and tat protein \(tat\) gene, partial cds](#)

5,060 bp linear RNA

Accession: JN687678.1 GI: 363584090

[Protein](#) [PubMed](#) [Taxonomy](#)

[GenBank](#) [FASTA](#) [Graphics](#)

3. [HIV-1 isolate BP00064_LH02 from USA gag protein \(gag\) gene, complete cds; pol protein \(pol\) gene, partial cds; vif protein \(vif\) and vpr protein \(vpr\) genes, complete cds; and tat protein \(tat\) gene, partial cds](#)

5,069 bp linear RNA

Accession: JN687679.1 GI: 363584096

[Protein](#) [PubMed](#) [Taxonomy](#)

[GenBank](#) [FASTA](#) [Graphics](#)

Revision date Custom range...

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Complete Record
Coding Sequences
Gene Features

Choose Destination
File Clipboard
Collections

Download 171 items.

Format FASTA

Sort by Default order

Show GI

Create File

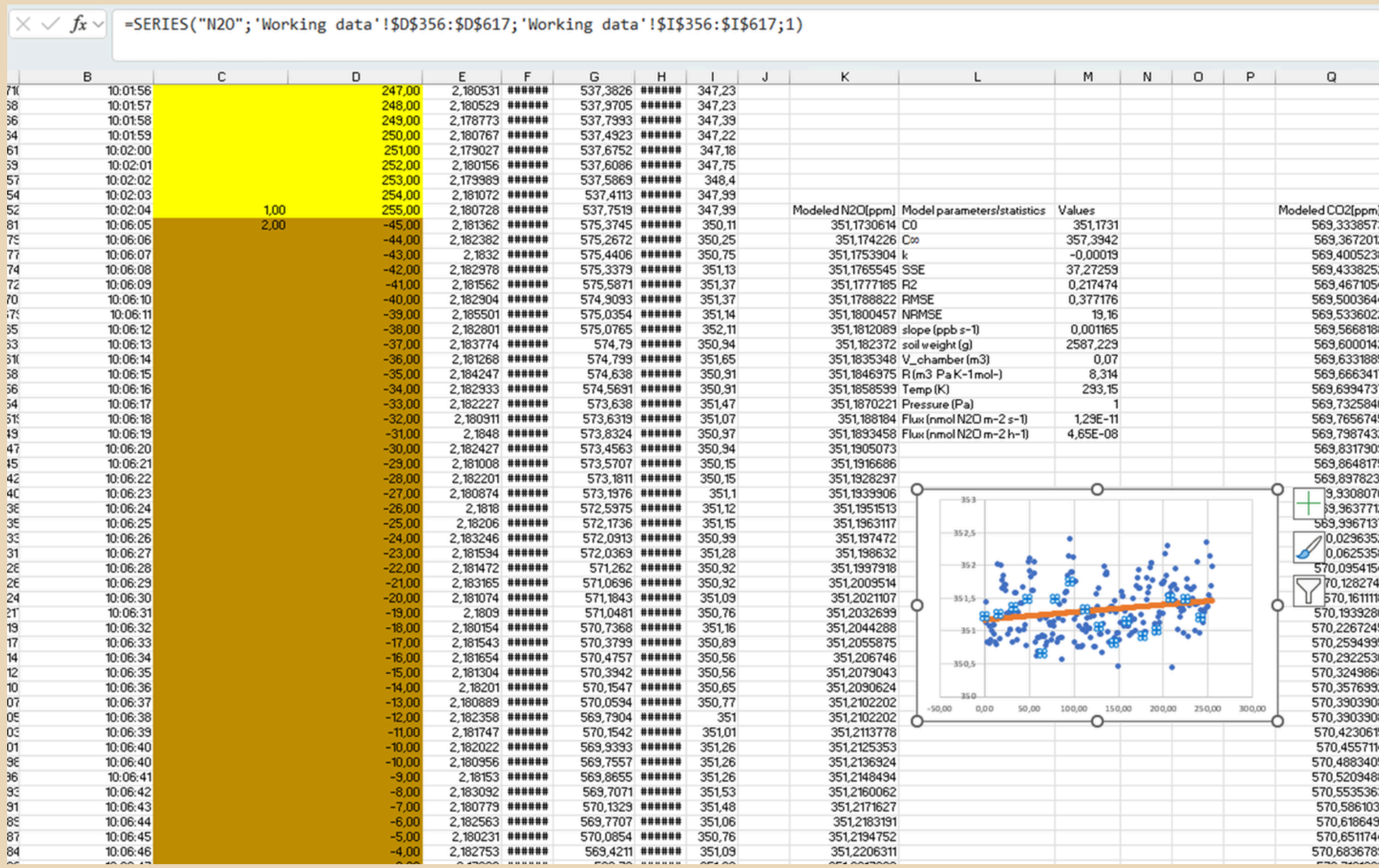
M12: Alignment Explorer (sequence_171.fasta)

Data Edit Search Alignment Web Sequencer Display Windows Help

DNA Sequences Translated Protein Sequences

Species/Abbrv	
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2. JN687678.1	GATGGGTGCGAGAGCGTCAGTATTAAGCGGGGGAGAAATTGGATAGATTGGGAGAAAAATTCGATTGAGGCCAGG
3. JN687679.1	GATGGGTGCGAGAGCGTCAGTATTAAGCGGGGGAGAAATTGGATAGATTGGGAGAAAAATTCGATTGAGGCCAGG
4. JN687680.1	GATGGGTGCGAGAGCGTCAGTATTAAGCGGGGGAGAAATTGGATAGATTGGGAGAAAAATTCGATTGAGGCCAGG
5. JN687681.1	GATGGGTGCGAGAGCGTCAGTATTAAGCGGGGGAGAAATTGGATAGATTGGGAGAAAAATTCGATTGAGGCCAGG
6. JN687682.1	GATGGGTGCGAGAGCGTCAGTATTAAGCGGGGGAGAAATTGGATAGATTGGGAGAAAAATTCGATTGAGGCCAGG
7. JN687683.1	GATGGGTGCGAGAGCGTCAGTATTAAGCGGGGGAGAAATTGGATAGATTGGGAGAAAAATTCGATTGAGGCCAGG
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9. JN687685.1	GATGGGTGCGAGAGCGTCAGTATTAAGCGGGGGAGAAATTGGATAGATTGGGAGAAAAATTCGATTGAGGCCAGG
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11. JN687687.1	GATGGGTGCGAGAGCGTCAGTATTAAGCGGGGGAGAAATTGGATAGATTGGGAGAAAAATTCGATTGAGGCCAGG
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25. JN687701.1	CAAGATAATACATACAGACAAATGGCAGCAATTTCAACAGTGCAGTAAAGGCAGCCGTTGGTGGGCAGG
26. JN687702.1	CAAGATAATACATACAGACAAATGGCAGCAATTTCAACAGTGCAGTAAAGGCAGCCGTTGGTGGGCAGG
27. JN687703.1	CAAGATAATACATACAGACAAATGGCAGCAATTTCAACAGTGCAGTAAAGGCAGCCGTTGGTGGGCAGG
28. JN687704.1	CAAGATAATACATACAGACAAATGGCAGCAATTTCAACAGTGCAGTAAAGGCAGCCGTTGGTGGGCAGG
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33. JN687709.1	CAAGATAATACATACAGACAAATGGCAGCAATTTCAACAGTGCAGTAAAGGCAGCCGTTGGTGGGCAGG
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35. JN687711.1	CAAGATAATACATACAGACAAATGGCAGCAATTTCAACAGTGCAGTAAAGGCAGCCGTTGGTGGGCAGG
36. JN687712.1	CAAGATAATACATACAGACAAATGGCAGCAATTTCAACAGTGCAGTAAAGGCAGCCGTTGGTGGGCAGG
37. JN687713.1	CAAGATAATACATACAGACAAATGGCAGCAATTTCAACAGTGCAGTAAAGGCAGCCGTTGGTGGGCAGG

sequence_171.fasta



Expectation

Steps to replicate the tables and figures

General information:

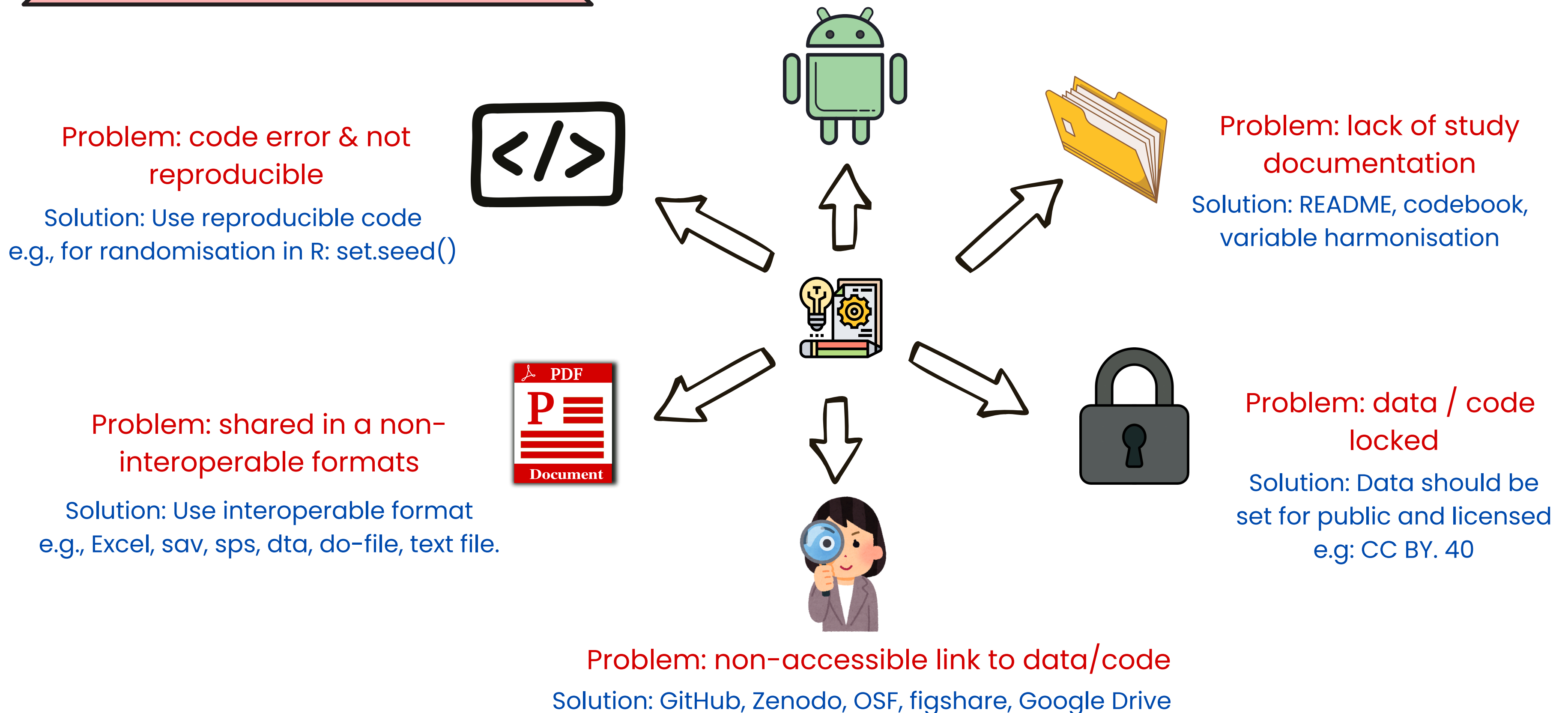
- Instructions for replication of the results using Stata. All do-files were created in Stata 16.
- There are 4 subfolders (DO-FILES, DTA-FILES, OUTPUT, XLS-FILES) in the replication_package folders. Copy these folders to your local computers and continue with the next step.

Manuscript: <https://doi.org/10.1016/j.crm.2021.100377>

Reality

migration-SLR / replication_package /	
IvoSteimanis Add files via upload	
Name	Last commit message
..	
DO-FILES	Add files via upload
DTA-FILES	Add files via upload
OUTPUT	Add files via upload
Readme.docx	Add files via upload
desktop.ini	Add files via upload
~\$Readme.docx	Add files via upload

Why data/code not reproducible? (and practical solutions)



OPEN SCIENCE CHECKLIST	
	Protocol
	Study registration
	Open materials
	Open data
	Open code
	Citing materials / data / code
	Code availability
	Reporting software & versioning
	Using reporting guidelines
	Open access
	Reporting the preprint

How to improve reproducibility?

Method reproducibility

Outcome reproducibility

Computational reproducibility

Inferential reproducibility

(*) Reproducibility definition by the EU

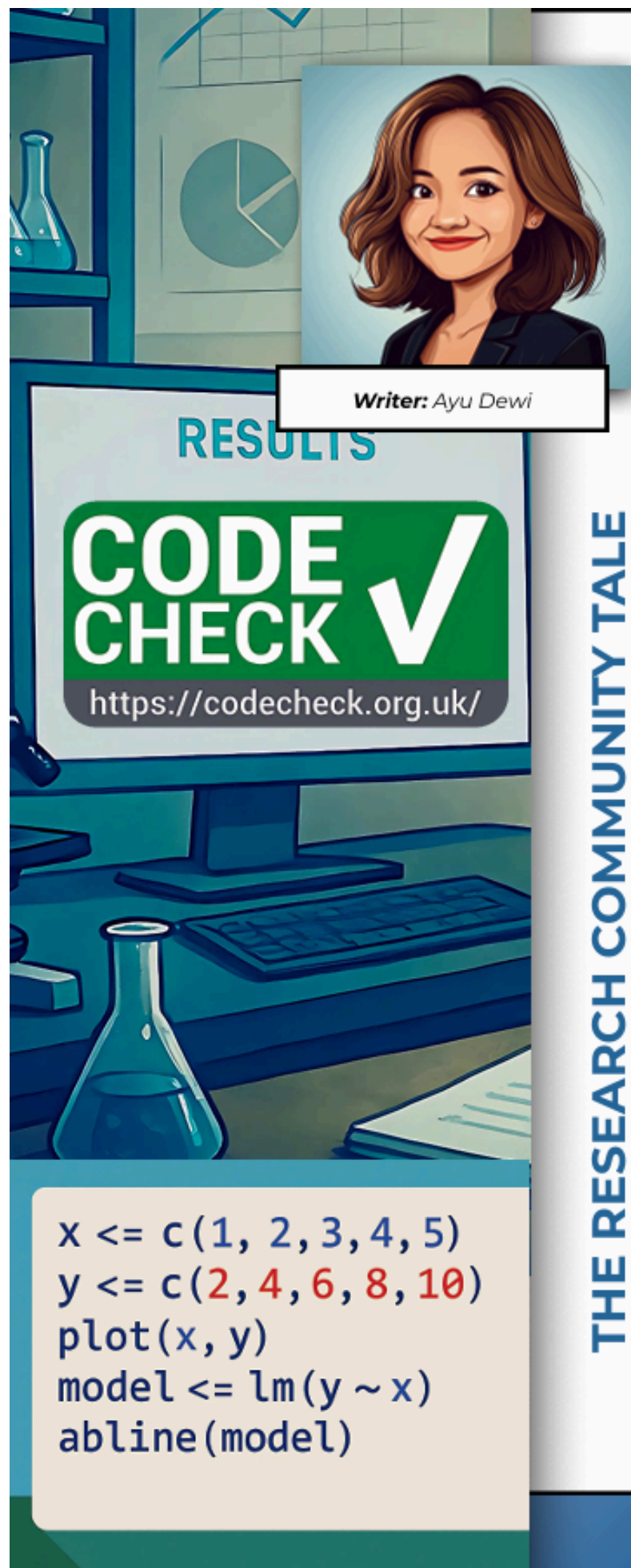
(*) *A Randomised Controlled Trial study is currently being conducted to test the effectiveness of an open science checklist in improving the reproducibility of research findings through an intervention implemented during the peer review of manuscripts submitted to biomedical journals.*

How to check reproducibility of my research?

- Ask peers to reproduce my data and analysis code, or
- Ask CODECHECK Amsterdam UMC: RDM Help-Desk.

2025-08 | Institutional CODECHECKs launched in the Netherlands

Two organisations in the Netherlands recently started offering CODECHECKs as an institutional service: the TU Delft Digital Competence Centre (DCC) together with 4TU.ResearchData and the Amsterdam University Medical Center (AUMC). Find out more on the new page for institutional CODECHECKs at <https://codecheck.org.uk/institutions> and read the blog post “CODECHECK in Practice: How TU Delft and 4TU.ResearchData Are Making Reproducibility Happen” (see also [on LinkedIn](#)) by the DCC team and the report “Computational reproducibility at Amsterdam UMC” by the AUMC team, which includes experiences from their first few checks and their future plans.



Writer: Ayu Dewi

03

ENHANCING REPRODUCIBILITY WITH CODECHECK AT AMSTERDAM UMC

Interviewee: Sam Langton, Research Software Consultant, Amsterdam UMC

Computational reproducibility, which ensures that the same data and analysis produce the same results, remains a major challenge in research across disciplines. CODECHECK² is a community initiative to solve this challenge through independent verification of computational workflows in scientific publications. This tale shows how CODECHECK is implemented at Amsterdam UMC (AUMC) to enhance reproducibility, address technical barriers and foster a culture of transparent research.

What is CODECHECK?

CODECHECK allows independent code checkers to rerun code, assess its reproducibility, and give constructive feedback. This process not only helps authors refine their documentation but also ensures that published findings are computationally sound.

CODECHECK pairs computational reviewers with subject-matter experts. It offers a dual-layer validation of research outputs such as tables, graphs and statistical analyses, to accurately reflect the results generated by the code. It is a free and open tool that operates independently of journals, allowing researchers to verify their research findings before the manuscript is submitted to a journal.

A successful code check earns a certificate as public endorsement of the study's reproducibility. It boosts credibility and showcases a commitment to open science.

CODECHECK is not limited to one field. While it has gained traction in neuroscience and geographic information systems, its reach is expanding. Fields such as medical sciences, social sciences and digital humanities are increasingly adopting the initiative.

Researchers around the world can incorporate CODECHECK into their research processes in a way that suits their needs. The system is designed to be flexible, allowing different types of studies to benefit from its reproducibility checks. Its adaptability makes it a standout choice for institutions looking to elevate their reproducibility frameworks.

Implementing CODECHECK at Amsterdam UMC

AUMC joined CODECHECK as part of a project focused on improving reproducibility by refining the management of research software and data. The initiative was built on previous successes, like the neuroscience reproducibility hackathon and collaborations with the Dutch Reproducibility Network. Unlike traditional audits, CODECHECK at AUMC was designed to identify practical barriers to reproducibility, such as IT infrastructure constraints and documentation gaps.

The process

CODECHECK at AUMC follows a structured approach.

- Researchers submit their code and data for verification
- An independent code checker reruns the code
- The code checker documents the process, identifies any discrepancies, and provides detailed feedback via a standard template
- If the research meets reproducibility standards, a CODECHECK certificate is issued, which can be published on platforms like Zenodo.

Training and support

Anyone can become a code checker after appropriate training. Founders Daniel Nüst and Stephen Eglen offer hands-on guidance, including one-on-one mentoring, to help researchers build the skills they need. While CODECHECK can serve as research support within institutions, it also invites PhD and master's students to conduct checks for their peers, fostering a collaborative learning

research support services. In future, researchers may access it through a ticketing system or as part of data management planning requirements. Embedding reproducibility checks into the research ecosystem aims to make them standard practice and foster a peer-to-peer verification culture.

Challenges

Despite its benefits, implementing CODECHECK at AUMC has revealed several challenges. It needs open data and code, when much research uses confidential or proprietary data. This complicates reproducibility checks. While anonymisation and synthetic data can sometimes provide solutions, institutional policies must support open data practices to facilitate broader adoption.

Running computational workflows requires access to appropriate IT infrastructure, including compatible software environments and sufficient computational resources.

Encouraging researchers to prioritise reproducibility remains an ongoing effort. Institutional incentives could counterbalance the perceived burden.

Future directions

For CODECHECK to have an impact, efforts must go beyond code checking alone. That means fostering a culture where researchers, especially early career researchers, verify each other's work, even beyond formal CODECHECK reviews. Integrating CODECHECK into institutional policies, such as requiring reproducibility checks for grants or research outputs, could further encourage adoption and ensure long-term viability.

Reflections

Reproducible research relies on two key factors: open data and open code, along with clear and thorough documentation. It's like giving someone a recipe with all the ingredients but leaving out the step-by-step instructions; without those details, they won't be able to recreate the dish exactly as you made it. The same applies to scientific research. Amsterdam UMC's adoption of CODECHECK is a step toward ensuring that studies are not just credible in theory but also verifiable in practice. By embracing initiatives like CODECHECK, we move closer to a future where transparency and reproducibility are at the heart of scientific discovery.



osiris4r.eu

Recommendations for the editor

17. Have somebody conduct a reproducibility check (e.g., via CODECHECK, <https://codecheck.org.uk>).
18. Let authors create a frozen version of the OSF project to prevent data from being altered or deleted or archive all materials via the journal.
19. Transition to a model with much lower and justified APCs or Diamond Open Access (for justified publication costs see <https://doi.org/10.12688/f1000research.27468.2>).

Reviewer's comment is publicly available :
<https://osf.io/pbnw7/files/p7rfw>

CODECHECK approved!

CODECHECK certificate 2024-004

10.5281/zenodo.13364677



Item	Value
Title	Open Science interventions to improve reproducibility and replicability of research: a scoping review preprint
Authors	Leonie Dudda, Eva Kormann, Magdalena Kozula, Nicholas J. DeVito, Thomas Klebe, Ayu P.M. Dewi, René Spijker, Inge Stegeman, Veerle Van den Eynden, Tony Ross-Hellauer, Mariska M.G. Leeftang

Open science debate



GDPR & patient inform consent

Data ownership

Open science is not universal

Familiarity with open technology

Preprints

Open access

L. Clinical Trials

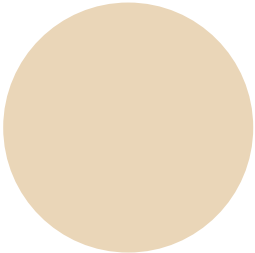
PAGE CONTENTS

1. [Registration](#)
2. [Authors' Access to Data](#)
3. [Data Sharing](#)

1. *Registration*

The ICMJE's clinical trial registration policy is detailed in a series of editorials (see [News and Editorials](#) and [FAQs](#)).

Briefly, the ICMJE recommends that all medical journal editors require registration of clinical trials in a public trials registry at or before the time of first patient consent for enrollment as a condition of consideration for publication.



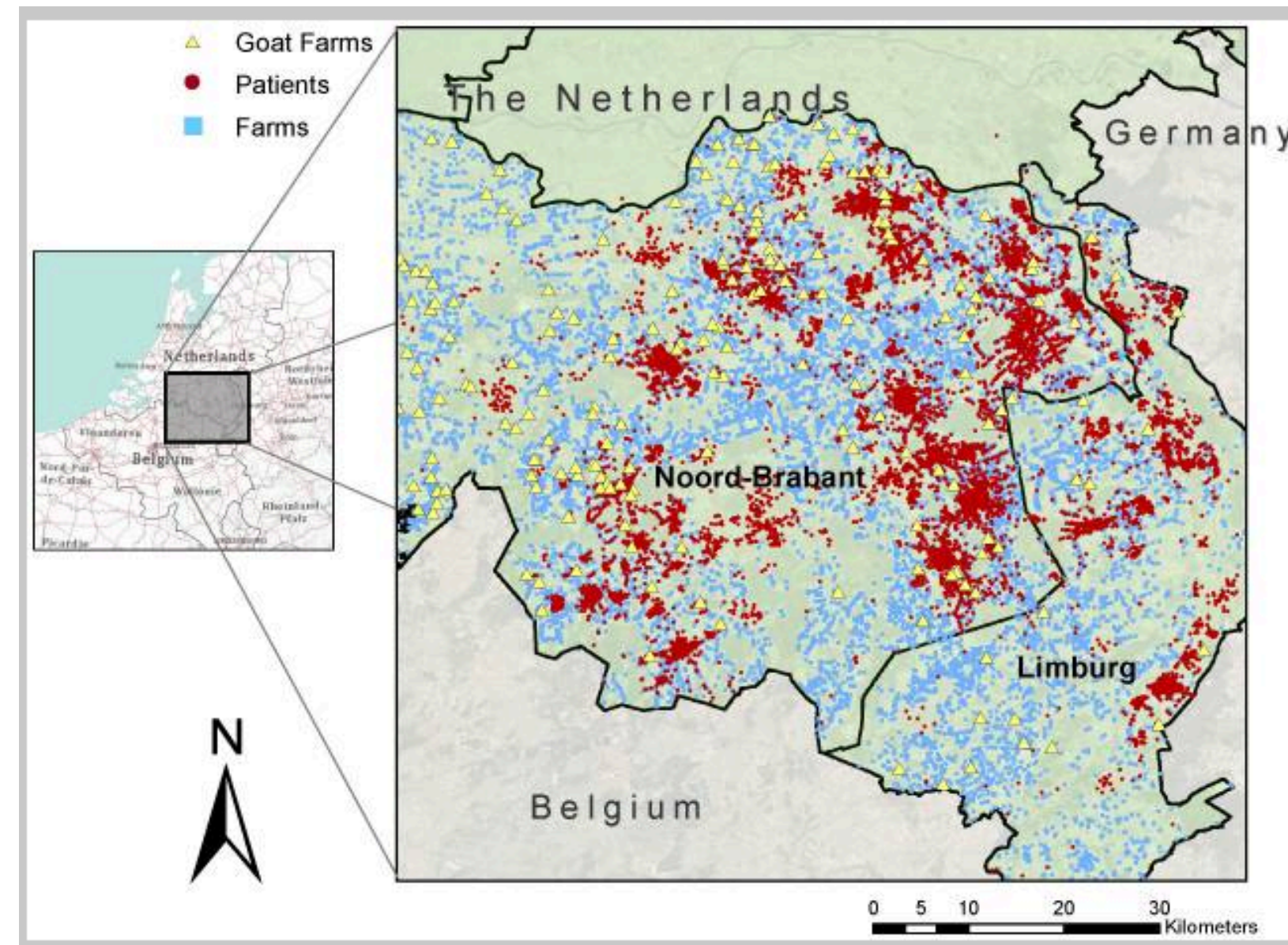
If you would like to read the full privacy notice of the RECOVERY study, it is available at www.recoverytrial.net/study-faq/data-privacy.

CONFIDENTIALITY:

- All information obtained during the course of this study, including hospital records, personal data and research data will be kept strictly confidential. Data that may be reported in scientific journals will not include any information that identifies you as a participant in this study.

Q Fever and Pneumonia in an Area with a High Livestock Density: A Large Population-Based Study

As open as possible,
as close as necessary!



Is this open science?



“OPEN SCIENCE FEELS HARD TO OPEN — UNTIL WE OPEN IT EVERY DAY. THEN IT’S JUST SCIENCE.”

“REPRODUCIBILITY ISN’T EXTRA WORK — IT’S UNFINISHED WORK DONE PROPERLY.”

Thank You!



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